Inbound Lead Triage

Agentic Specialist Capstone
Automating first-pass lead qualification with Zapier, Slack, Gmail, and Python

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Problem, Goal & Success Criteria

Problem: inbound leads require repetitive reading, extraction, qualification, prioritization, routing, and follow-up.

Goal: automate first-pass triage while preserving human review and control of external communication.

Baseline assumption: 100 leads/month, 10 minutes/lead, $40/hour fully loaded labor cost.

Acceptance targets: 100% regression pass rate; >=3 representative runs; invalid data must be safely flagged; projected review time <=2 minutes/lead.

End-to-End Architecture

Gmail -> Orchestrator Zap -> Slack + Zapier Table

Slack NEW LEAD -> Python qualification worker -> structured thread result

Slack -> Qualification & Routing Zap -> routing recommendation

Human review -> Reply Sender Zap -> Gmail

Slack acts as the shared operational state; human review remains available for exceptions and external responses.

Orchestration Blueprint

Orchestrator Zap: classifies inbound email, posts structured work to Slack, and logs the record.

Python Worker: validates structured lead fields and performs deterministic qualification logic.

Qualification & Routing Zap: supports lead-quality, urgency, and routing actions.

Reply Sender Zap: handles approved external email responses.

Coordination model: hybrid no-code + Python, using Slack threads as shared state and handoff points.

Python Agent - Capstone-Aligned V2

Deterministic inputs: company size, use case, industry, urgency signals.

Scoring weights: Company Size 35, Use Case 25, Industry 20, Urgency 20 = 100 total.

Outputs: lead_quality_score, ICP fit assessment, suggested response priority, validation status/issues, scoring reason.

Example: Example SaaS Company -> 95/100 -> Strong Fit -> High Priority.

Missing required information returns an invalid result instead of a guessed score.

Prompt Engineering & Safety

Base prompt uses fixed intent categories: DEMO REQUEST, PRICING QUESTION, PARTNERSHIP, GENERAL INQUIRY.

Meta-prompt review targeted ambiguity, unsupported assumptions, output inconsistency, and unsafe missing-data handling.

Version 1.1 added a defined fallback, no-invention rule, structured output, and sensitive-data boundaries.

Fallbacks: unclear intent -> GENERAL INQUIRY; missing qualification data -> invalid/human review; automation failure -> manual Slack review.

Prompt and orchestration changes are version-tracked and regression-tested.

Quality Specification

Quality dimensions:
Correctness, validation, consistency, explainability, human oversight, speed, and cost.

Scoring scale:
5 = fully correct / no correction needed
4 = correct with minor issue
3 = acceptable / minor human review needed
2 = significant error
1 = failed / unusable

Manual quality baseline:
Representative leads are manually reviewed for correct qualification and routing, then compared with automated results.

Passing standard:
Average score >= 4/5; no dimension below 3/5; correct qualification and routing; 100% regression tests pass.

Evaluation Criteria & Representative Runs

Working Slack prototype run 1: Qualified lead -> 75 -> PASS.

Working Slack prototype run 2: Needs Review -> 50 -> PASS.

Working Slack prototype run 3: Not Qualified -> 18 -> PASS.

Working Slack prototype run 4: Missing budget -> invalid, score 0 -> PASS.

These runs demonstrate normal and incomplete-input handling in the Slack workflow.

Regression / Harness Evidence

Capstone-aligned V2 pytest suite collected 4 tests.

Strong Fit / High Priority: PASS.

Moderate Fit / Medium Priority: PASS.

Low Fit / Low Priority: PASS.

Missing required field / invalid input: PASS.

Final result: 4 passed in 0.05s - 100% pass rate.

Performance Results & Business Value

Current-state assumption: 100 leads/month x 10 min = 16.67 human hours/month.

Estimated current labor cost: ~$667/month (~$8,004/year).

Projected state: ~2 minutes human review/lead = ~3.33 hours/month.

Projected labor cost: ~$133/month; savings ~$534/month or ~$6,408/year.

Projected first-pass processing-time reduction: 80%. These are planning projections to validate in a controlled pilot.

Governance, Deployment & Adoption

Human review remains available for ambiguous, invalid, and unexpected results.

Credentials stay outside source code; synthetic data is used for demonstration evidence.

Monitor worker connection failures, invalid data, unexpected scores, failed Zaps, API errors, and regression failures.

Rollback: stop automation -> manual Slack review -> diagnose/fix -> run regression suite -> resume only after validation.

Scale plan: centralized logging/secrets, retries, alerting, production integrations, and measured operational KPIs.

Conclusion / Capstone Emphasis

Demonstrates an end-to-end hybrid agentic workflow across Gmail, Zapier, Slack, and Python.

Python V2 provides deterministic lead quality scoring, ICP fit assessment, and response priority.

Evidence includes prompt iteration, quality criteria, representative runs, regression testing, governance, and rollback.

Business case projects ~80% reduction in first-pass human effort and ~$6,408 annual OPEX savings under stated assumptions.

Status: READY FOR CAPSTONE DEMONSTRATION and controlled pilot evaluation.